Editorial

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Standards in systems and synthetic biology: status, developments, and tools in 2026

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Abstract: For over a decade, the Journal of Integrative Bioinformatics has published special issues providing updated specifications for standards in systems biology and synthetic biology. Beginning in 2024, the special issues expanded to include novel or refined tools that leverage these standards or play a key role in their implementation. The current special issue contains four updated specifications and two tools or libraries. In addition, this document functions as a comprehensive reference for all COMBINE standard specifications, ensuring researchers and practitioners can access the most up-to-date versions available in the field.

1 Introduction

The "COmputational Modeling in BIology NEtwork" initiative (COMBINE, see https://co.mbine.org/) seeks to harmonise the development of various community standards for computational models in biology [1, 2]. It coordinates and supports related projects to build a cohesive set of compatible, interoperable, and comprehensive standards spanning the entire landscape of modelling in systems and synthetic biology. An overview of COMBINE standards and associated initiatives is presented in Figure 1. Since 2015, periodic special issues dedicated to COMBINE standards have been published, documenting updates over the years [3–10].

This special issue contains the most recent updates to the following COMBINE standards:

1. SBML level 3 package: Flux Balance Constraints version 3 [11] for constraint-based biological model encoding, adds user-defined constraints, quadratic variables, flexible species charges and generic formula

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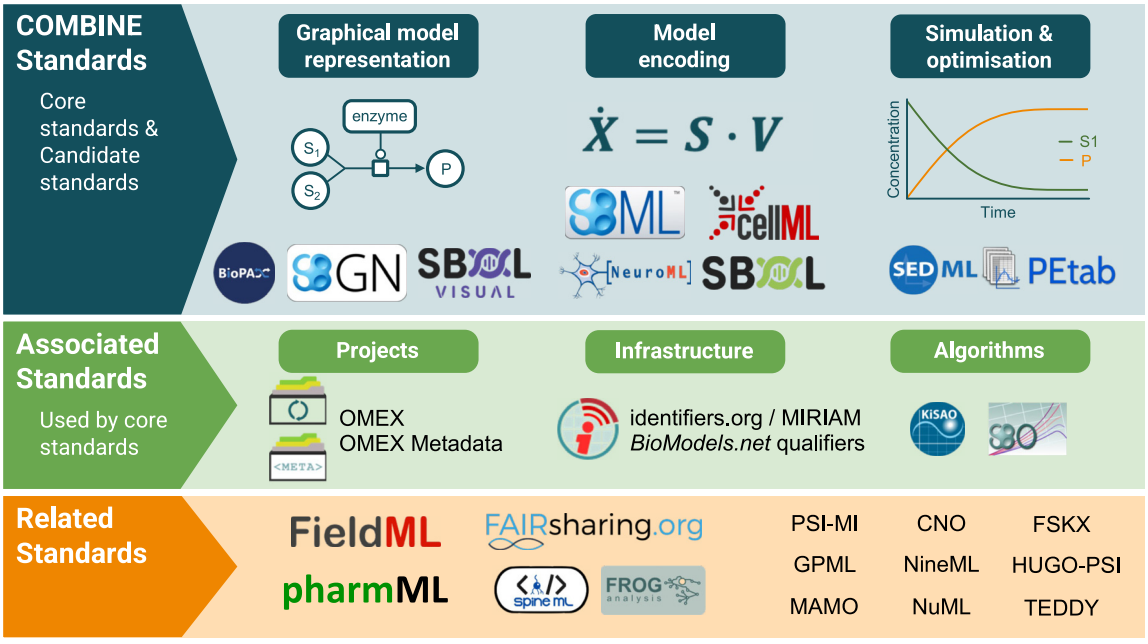


Figure 1: COMBINE standards and associated efforts (image adapted from [3]).

components while maintaining backward compatibility and enabling interoperable cross-community model exchange.

2. SBGN Process Description Language level 1 version 2.1 [12] for visualising biological pathways focusing on event order using entity pools and processes with defined edges for interactions, version 2.1 refines terminology, updates content and revises figures.
3. SBGN Activity Flow Language level 1 version 1.3 [13] for biological pathway visualisation focusing on biological activities, version 1.3 provides refined specifications that improve consistency, terminology and interpretability.
4. The Parameter Estimation tables (PEtab) format version 2 [14] for parameter estimation in systems biology and medicine, version 2 simplifies problem definitions and extends support for complex experimental setups and diverse model formats.

This special issue also presents some software, libraries and tools that are utilising these standards or are important to their implementation. Tool overviews can be found on the webpages of the respective standards, new or updated tools and libraries presented here are:

– SBOL Converter [15], a backward-compatibility tool that facilitates transitions between SBOL versions, validates designs against compliance rules, and processes other biological formats such as GenBank and FASTA.
– A standardised SBML/PRISM benchmark library for stochastic model checking in synthetic biology [16], a curated library of biologically inspired stochastic models in both SBML and PRISM to improve reproducibility, enable tool comparisons and accelerate software infrastructure development.

2 Current versions of COMBINE standards

In this issue, similar to our past special issues, we will provide a concise summary of all COMBINE standards. For the most current specifications of COMBINE standards, please consult the sections that follow. To facilitate ease of navigation, the structure and core information remain consistent with previous special issues.

2.1 Core standards

A comprehensive overview about the COMBINE core standards, that can be interactively searched and browsed, can be found as COMBINE collection in the FAIRsharing platform (https://fairsharing.org/3495) as well as at the COMBINE website (https://co.mbine.org/). The COMBINE core standards are also referenced by several standards published by the International Organization for Standardization (ISO). ISO 20691:2022 "Biotechnology – Requirements for data formatting and description in the life sciences" (https://www.iso.org/standard/68848.html) provides recommendations and requirements for the model formatting, as well as for the semantic description and annotation of data and models in the life sciences, and recommends the COMBINE core standards in its annex. ISO/TS 9491-1:2023 "Biotechnology – Recommendations and requirements for predictive computational models in personalised medicine research – Part 1: Guidelines for constructing, verifying and validating models" (https://www.iso.org/standard/83516.html) specifies requirements and recommendations for models used for research purposes in the field of personalised medicine and provides guidelines to apply the COMBINE core standards in that field.

2.1.1 BioPAX (biological PAthway eXchange)

BioPAX is a language designed for the integration, exchange, and analysis of biological pathway data. BioPAX is represented using Web Ontology Language (OWL). The current specification is:

Standard	Specification	Reference
BioPAX [17]	BioPAX	[18]

Tools for BioPAX include Paxtools [19], PathVisio [20], and ChiBE [21].

2.1.2 CellML

The CellML language is an XML-based markup language designed for the storage and exchange of computer-based mathematical models. The current specifications are:

Standard	Specification	Reference
CellML [22]	CellML 2.0.1	[23]
	CellML metadata framework 2.0	[24]

The CellML Metadata Framework [24] is now deprecated in favour of the OMEX Metadata Specification [25]. Tools for CellML include libCellML (https://libcellml.org) and OpenCOR [26]. A tool overview can be found at https://cellml.org/tools.

2.1.3 NeuroML

The Neural Open Markup Language (NeuroML) is a description language based on XML, offering a standardised data format for the definition and exchange of neuronal cell and network model descriptions. The current specification is:

Standard	Specification	Reference
NeuroML [27, 28]	NeuroML version 2.3	[28]

Tools for NeuroML include jNeuroML [28], NetPyNE [29], and EDEN [30]. A tool overview can be found at https://docs.neuroml.org/Userdocs/Software/Software. The development and growth of the NeuroML ecosystem of tools was recently reviewed in [27].

2.1.4 SBGN (Systems Biology Graphical Notation)

The Systems Biology Graphical Notation (SBGN) provides a suite of standardised graphical languages designed for visually representing biological knowledge. It encompasses three distinct languages, Process Description, Entity Relationship, and Activity Flow. Furthermore, SBGN-ML, an XML-based file format, is utilised for detailing the geometry of SBGN maps. The current specifications are:

Standard	Specification	Reference
SBGN [31]	SBGN process description level 1 version 2.1	[12]
	SBGN entity relationship level 1 version 2.0	[32]
	SBGN activity flow level 1 version 1.3	[13]
	SBGN markup language version 0.3	[33]

Tools for creating SBGN maps include Newt [34], CySBGN [35], PathVisio (SBGN plugin) [20], and SBGN-ED [36]. SBGN support can be integrated into other software tools using cytoscape.js [37]. A tool overview can be found at https://sbgn.github.io/, as well as in [38].

2.1.5 SBML (Systems Biology Markup Language)

The Systems Biology Markup Language (SBML) [39, 40] is an XML-based format designed for computer interpretation of models of biological processes. While it is especially suited for models that describe processes, SBML's application is not confined to these alone. The current specifications are:

Standard	Specification	Reference
SBML [41]	SBML level 3 core, version 2, release 2	[39]
	SBML level 3 package: distributions, version 1, release 1	[42]
	SBML level 3 package: flux balance constraints version 3	[11]
	SBML level 3 package: groups, version 1, release 1	[43]
	SBML level 3 package: hierarchical model composition, version 1, release 3	[44]
	SBML level 3 package: layout, version 1, release 1	[45]
	SBML level 3 package: multistate, multicomponent and multicompartment species, version 1, release 2	[46]
	SBML level 3 package: spatial processes, version 1, release 1	[47]
	SBML level 3 package: qualitative models, version 1, release 1	[48]
	SBML level 3 package: render, version 1, release 1, release 1	[49]

Tools for SBML include COPASI [50], roadrunner [51], sbmlutils [52] and CySBML [53] (https://sbml4humans.de). A tool and model overview can be found at https://sbml.org/software/.

2.1.6 SBOL (Synthetic Biology Open Language)

The Synthetic Biology Open Language (SBOL) is a language for detailing and sharing information about biological components, devices, and systems. SBOL Visual (SBOLv), a related standard, offers a uniform collection of symbols and guidelines for illustrating genetic circuits. The current specifications are:

Standard	Specification	Reference
SBOL [54]	SBOL version 3.1.0	[55]
	SBOL visual version 2.3	[56]
	SBOL visual version 3.0	[57]

Tools for SBOL and SBOL Visual include SynBioHub [58], SBOLCanvas [59], paraSBOLv [60] and VisBOL [61]. A tool overview can be found at https://sbolstandard.org, as well as in [62].

2.1.7 SED-ML (Simulation Experiment Description Markup Language)

The Simulation Experiment Description Markup Language (SED-ML) is a format based on XML that is used for detailing simulation experiments. It enables the specification of which models to use, the experimental tasks to execute, and the results to generate. SED-ML supports models that are encoded in a variety of languages. The current specification is:

Standard	Specification	Reference
SED-ML [63]	SED-ML level 1 version 5	[64]

Tools for SED-ML include COPASI [50] and many of the tools listed on BioSimulators (https://biosimulators.org/). A tool overview can be found at https://sed-ml.org/showcase.html.

2.2 Associated standards

Associated standards provide an additional layer of semantics to COMBINE representation formats. The current specifications are:

Standard	Specification	Reference
COMBINE archive [65]	COMBINE archive 1.0	[66]
OMEX metadata	OMEX metadata version 1.2	[25]
BioModels.net qualifiers [67]	–	[68]
Identifiers.org URIs [69]	–	[70]
Systems biology ontology [71]	[External] bioportal	[72]
Kinetic simulation algorithm ontology [71]	[External] bioportal	[73]
Parameter estimation tables (PEtab)	PEtab format version 2	[14]

A COMBINE archive consolidates multiple documents and essential information required for a modelling and simulation project into a single file. This archive utilises the Open Modeling EXchange (OMEX) format for encoding. The COMBINE archive metadata offers a unified, community-endorsed method for annotating diverse standardised model and data formats contained within a COMBINE archive.

BioModels.net qualifiers represent standardised relationships (predicates) that define the connection between an object in a descriptive language and the external resource used for its annotation. MIRIAM Unique Resource Identifiers (URIs) enable the unique and unambiguous identification of an entity in a consistent and lasting way. The MIRIAM Registry offers a set of services and resources that assist in creating, understanding, and resolving MIRIAM URIs. Using Identifiers.org technology, MIRIAM URIs can be accessed in a versatile and reliable manner. These URIs are used by controlled annotation schemes in SBML, SED-ML, CellML, SBOL, and BioPAX.

The Systems Biology Ontology (SBO) comprises a collection of controlled, relational vocabularies encompassing terms frequently used in Systems Biology, especially within the realm of computational modelling. Every

component within an SBML (Systems Biology Markup Language) file may include an optional attribute named sboTerm, which should correspond to a specific term from the SBO. Furthermore, every symbol used in SBGN (Systems Biology Graphical Notation) is linked to an appropriate term from the SBO. SBOL also utilises SBO to specify types of genetic parts and molecular interactions.

The Kinetic Simulation Algorithm Ontology (KiSAO) describes various algorithms along with their characteristics and the relationships between them through their specific features and parameters. It is utilised within the Simulation Experiment Description Markup Language (SED-ML), enabling simulation software to automatically select the optimal algorithm for a given simulation and unambiguously refer to it.

The OMEX Metadata Specification serves as a technical implementation of the community consensus among COMBINE standards, aimed at standardising the description of computational models and other resources through metadata, as outlined by [74].

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